Supplementary Material for

Surrogate Selection for Reinforcement-Learning HVAC Control: Step Size, Not Predictive Accuracy, Predicts Control Utility

Supplementary Material

The main article reports every numerical value needed to evaluate the surrogate-selection claim and the role-separated architecture, but not the reproduction detail behind them. This Supplement carries that detail. It collects the full nomenclature, the pre-specified hypotheses and their verdicts, the complete limitations statement, implementation formulas, calibration and hyperparameter tables, seed-robustness and per-regime summaries, and the supporting diagnostics for each part of the study. The full project codebase is not publicly released at this stage; the building emulators are provided by the open-source BOPTEST framework (Blum et al., 2021; `https://github.com/ibpsa/project1-boptest`) and are deployed as a Docker container as described in the run-time-environment section below. All schematic figures are conceptual diagrams and are not generated from numerical datasets unless stated otherwise.

How to read this Supplement. Section 1 maps each main-article section to the supplementary items that support it. Sections 2 and 3 carry the pre-specified hypotheses and the full limitations statement. Section 4 is the bulk: supporting figures and tables organised in three parts mirroring the main article's evidence (predictive validation; controller utility and mechanism; transfer boundary), followed by the run-time environment, implementation-level formulas and additional tables.

Notation (read first). This Supplement uses the same names as the main article: **BB** for the compact black-box surrogate and **GB** for the calibrated grey-box twin. Two figures, the BB learning curve (Figure S1) and the warm-start negative control (Figure S10), still carry the legacy implementation labels `v3` and `v3.5` inside the plotted artwork; there `v3` is BB and `v3.5` is GB.

Supplementary Nomenclature

Symbol	Unit	Meaning
Thermophysical quantities		
T_{zone}	°C	zone air temperature
T_{amb}	°C	ambient air temperature
T_{sup}	°C	supply-air temperature command
C_{zon}	$\mathrm{J\,K^{-1}}$	zone thermal capacitance
R	$\mathrm{K\,W^{-1}}$	lumped envelope thermal resistance
η	–	heating power conversion efficiency
$\dot{Q}$	W	net zone heat flow
P	W	total HVAC electrical power
Δt	s	control / integration step (900 s)
Reinforcement-learning formulation		
s_t, a_t, o_t, d_t	–	state, action, observation, disturbance at step t

Symbol	Unit	Meaning
Φ	–	one-step plant (emulator) transition map
f_θ, g_ϕ	–	black-box surrogate (BB) and frozen physical twin (GB)
r_t	–	per-step (shaped) reward
$\tilde{r}_t$	–	hybrid reward with disagreement censor
$J(\pi)$	–	expected discounted return
γ	–	discount factor
w_c, w_e, w_s	–	comfort / energy / safety preference weights
$\hat{A}_t$	–	generalized advantage estimate
ρ_t	–	PPO probability ratio
ϵ	–	PPO clipping parameter
δ	–	surrogate–twin disagreement penalty
λ_T, λ_P	–	hybrid temperature / power censor weights
Evaluation metrics		
m_s	–	operational maintenance score evaluated on BOPTEST (lower is better)
r_{time}, r_{sev}	–	time-in-violation and severity components of m_s
$m_{s,\text{RL}}$, $m_{s,\text{PI}}$	–	operational score of the RL and built-in PI controller
RMSE_T	°C	live closed-loop zone-temperature RMSE
CV(RMSE)	%	coefficient of variation of the RMSE
NMBE	%	normalized mean bias error
R^2	–	coefficient of determination
Violation	%	fraction of steps outside the comfort band
CV	–	coefficient of variation (std/mean) over seeds
g_a	–	L_2 action-gap norm (surrogate vs live)
Δm_s	–	live-minus-surrogate m_s transfer gap
Transferability (testcase k)		
τ	–	pre-specified pass threshold $1.25\, m_{s,\text{PI}}$
ΔE	%	energy change of the RL controller vs PI
G_{RMSE}	%	rollout-RMSE improvement after Stage A/B/C recalibration
ρ_C	–	re-identified C_{zon} ratio vs the `bestest_air` baseline
$\mathcal{A}_k$	–	pre-specified actuator adapter
Abbreviations		
BOPTEST	–	Building Optimization Testing Framework
RTE	–	(BOPTEST) Run-Time Environment
DRL / PPO	–	deep RL / proximal policy optimization
GAE	–	generalized advantage estimation
HDRL	–	hierarchical deep reinforcement learning
MORL	–	multi-objective reinforcement learning
POMDP	–	partially observable Markov decision process
RC / ODE	–	resistance–capacitance / ordinary differential equation
PI / RBC / MPC	–	proportional–integral / rule-based / model predictive control

1 Map to the main article

Main-article section	Supporting material in this Supplement
§1 Introduction	Pre-specified hypotheses H1–H5 and their verdicts (§2)
§3.1–3.2 Test case, control problem, metrics	Full comfort-reward form (Table S14); 17D observation breakdown (Table S13); evaluation scenarios (Table S11); test-case list (Table S23); BOPTEST RTE loop (Figure S25)
§3.3 Surrogate backends	Data corpora (Table S2); architectures (Table S3); BB configuration and scaling (Tables S4, S5); Stage A/B/C calibration (Tables S6–S8, S34); learning curve and physical-consistency audit (Figures S1, S2); modeling assumptions and transition schema (Tables S32, S33)
§3.4 Role-separated hybrid	Mechanism schematic (Figure S6); censor-weight sweep (Table S17); censor formula (§6)
§3.5 Controllers and protocol	Controller-learning pipeline (Figure S4); PPO hyperparameters (Table S12); runtime benchmark (Table S10, Figure S5)
§4.1 Predictive fidelity	Multi-horizon rollout errors (Table S9); episode replicability (Figure S3); corpus-matched attribution waterfall (Figure S27)
§4.2 Fidelity–utility paradox	Live KPI breakdown (Figure S7); three-seed band (Table S15, Figure S9); maintenance-score decomposition (Table S30); controller-learning ledger (Table S31)
§4.3 Mechanism	Response-surface sharpness (Table S16); action-density phase portrait (Figure S8)
§4.4 Role separation restores control	Warm-start negative control (Table S18, Figure S10); transfer-gap diagnostics (Table S19, Figure S11)
§4.5–4.6 Boundary tests	HDRL architecture and sweep (Figures S12, S13, S14, Table S20); MORL pipeline and seeds (Figures S15, S16, S17, Tables S21, S22); transfer protocol, adapters and regimes (Figures S19, S20, S21, Tables S24, S25); identified physics and capacitance consistency (Tables S26, S28); hypothesis closure (Tables S35, S29, Figure S24)
§4.7 Limitations	Full limitations statement (§3)

2 Pre-specified hypotheses

Five claims were stated in advance of the corresponding runs and are reported whether supported or falsified. H1–H2 are the primary paradox and its remedy; H3–H5 are boundary tests that map the recipe's scope. H5 was registered after the first four, once the inversion itself had been established, and its first execution was invalidated by a defect in the planner rather than by the data; that run is recorded rather than discarded silently.

H1. *Fidelity–utility:* a surrogate with higher predictive fidelity is a better RL training environment. **Falsified.**

H2. *Role separation:* using the physical twin as a frozen reward-shaping censor over a smooth black-box rollout recovers the control utility that direct use of the twin destroys. **Supported.**

H3. *Censor-weight universality:* a single model-disagreement censor weight is sufficient across controller families. **Falsified**, which is what establishes controller-family specificity.

H4. *Transferability:* the inverse-calibration pipeline and the frozen controller transfer to related hydronic testcases under documented recalibration. **Partially supported:** the pipeline transfers, the frozen controller does not.

H5. *Controller-class specificity:* the fidelity–utility inversion belongs to policy-gradient learning, so a receding-horizon planner over the same surrogates escapes it. **Falsified:** the planner reproduces the inversion, reaching $m_s = 0.547/0.869$ on the least accurate backend against 1.051/1.369 on the most accurate one, within 0.005 of the learner on the peak window.

Stating these in advance is what lets the negative outcomes be read as findings rather than tuning artifacts. At the controller-experiment level the same discipline is applied through the CL1–CL5 ledger of Table S31; at the transfer level through Tables S35 and S29.

3 Full limitations statement

The main article reports an abridged limitations paragraph. The complete statement is as follows. Evaluation is simulation-only on a single-zone `bestest_air` emulator, with multi-zone and richer topologies untested and transfer limited to a hydronic family. First, the fidelity–utility relationship is established at two directionally sound operating points (hourly BB and the calibrated GB) together with the Δt-rescaling control, rather than as a continuous curve. The matched-resolution backend is *excluded* from the mechanism argument: an audit of its response sign finds it inverted on the supply-temperature command in about nine sampled states out of ten, reproducibly across four independent training draws while its 24 h rollout error improves, and inverted on the fan command as well. A training constraint that repairs the supply channel relocates the defect to the fan channel rather than removing it, so that backend is reported as a failure of the predictive criterion, not as a data point about temporal resolution. The seed replication ($N = 3$, seeds $\{42, 43, 44\}$) shows the remaining verdicts are seed-stable (Table S15), but the fidelity axis is still sampled at two sound points, and an intermediate-resolution sweep (e.g. 1800 s) on a surrogate audited across all actuated inputs is left to future work. Second, controller utility is established against two references: BOPTEST's built-in PI ($m_s = 0.910$) and a pre-registered receding-horizon MPC planning through the same surrogates. The MPC is the stronger of the two ($m_s = 0.547/0.869$ on the coarse BB backend) but it is a fixed, untuned configuration rather than a tuned Guideline-36 or industrial controller: its comfort violation is still 34.5% and 55.6% on the two windows. It was deliberately not tuned against m_s, the metric it is scored on. The planner is also a first-order method through the differentiable surrogate, so the comparison separates gradient-based optimization from other synthesis routes rather than planning from learning; a derivative-free planner is left to future work. Third, the surrogate power channel is reproduced less accurately than temperature (calibrated power-head MAE 482 W, about 9% of $P_{\max} = 5500$ W, rising to ∼2.3 kW in the partial-calibration diagnostic regime); the reported energy KPIs (Figure S7) and the transfer ΔE are live-versus-live BOPTEST measurements and do not inherit this surrogate error, but the weaker power channel limits predictive-fidelity claims about the power head and adds noise to the training-time energy reward, and a cumulative surrogate-versus-BOPTEST energy balance for at least one window is left to future work. Fourth, the controller-family specificity of the censor weight that falsifies H3 rests on a non-uniform sweep density (a dense four-point λ_T sweep for HDRL but only two measured endpoints for thermostatic PPO and a single adopted configuration for MORL), so it is read as consistent with family-specific regularization rather than as a fully-swept per-family optimum. Finally, the paradox mechanism is *measured* statically on each surrogate's one-step action map and corroborated by the Δt-rescaling control, which preserves response signs by construction, rather than by the matched-resolution ablation, but a loss-landscape analysis along the PPO trajectory and generality to off-policy algorithms (SAC, TD3) remain open; statistical support otherwise varies by family (MORL $N = 5$, neutral $m_s = 0.187 \pm 0.078$ with CI far below the PI 0.910; HDRL $N = 3$ at each of four swept λ_T), and the backend×observation grid is only partially crossed.

4 Supporting diagnostics

Supplementary figures and tables

The following figures and tables provide supporting schematics, calibration and controller diagnostics, and detailed per-regime and per-seed breakdowns. They are cited from the main text (as Supplementary Figure/Table references) but are not required to follow the main argument.

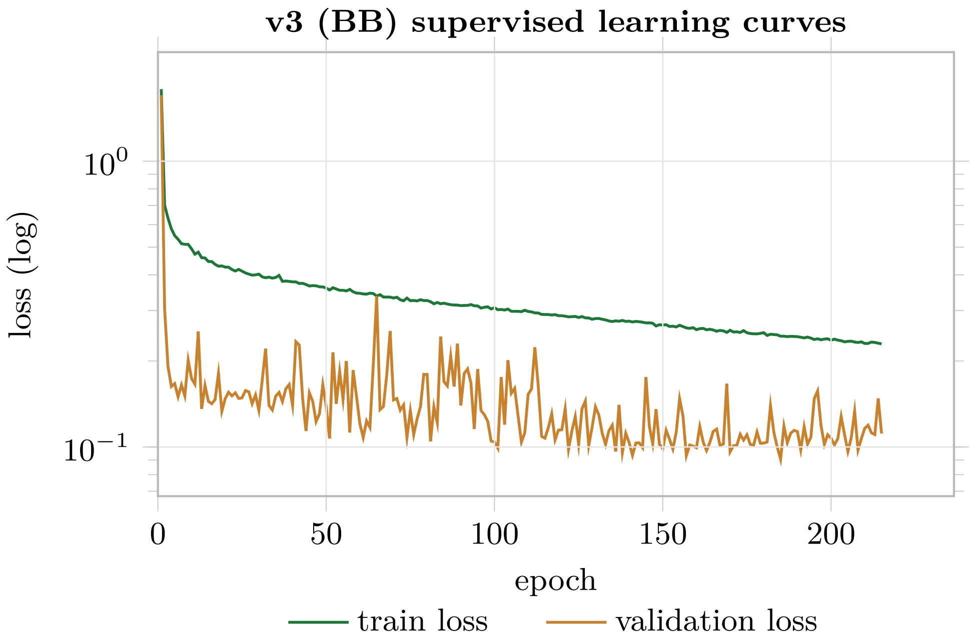


Figure S1: BB supervised train/validation learning curves. Early stopping at the best validation epoch precedes any validation-loss upturn, indicating no overfitting at the selected checkpoint.

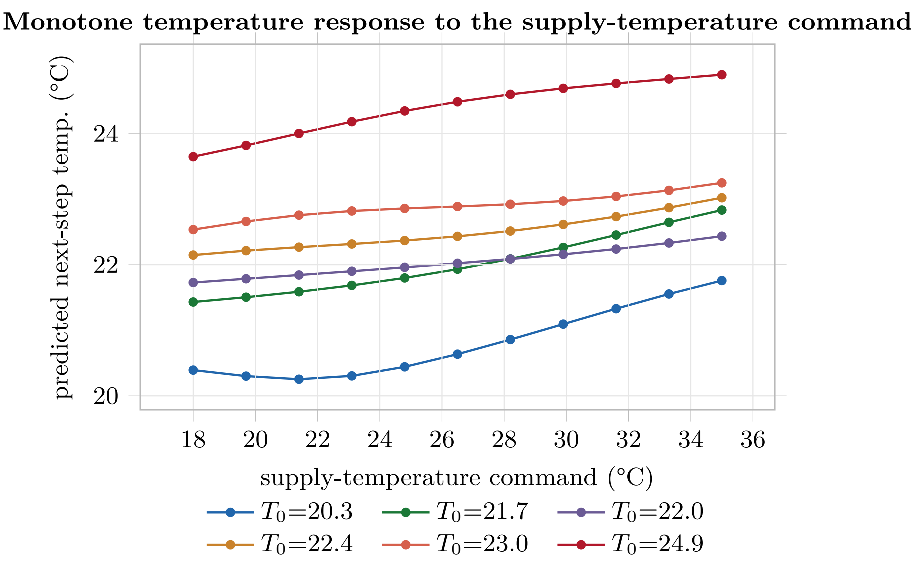


Figure S2: Physical-consistency audit: predicted next-step zone temperature versus the supply-temperature command for representative initial states. The response has the correct control sign and is predominantly increasing (mean Spearman 0.91), with small departures from strict monotonicity contributed by the neural residual head.

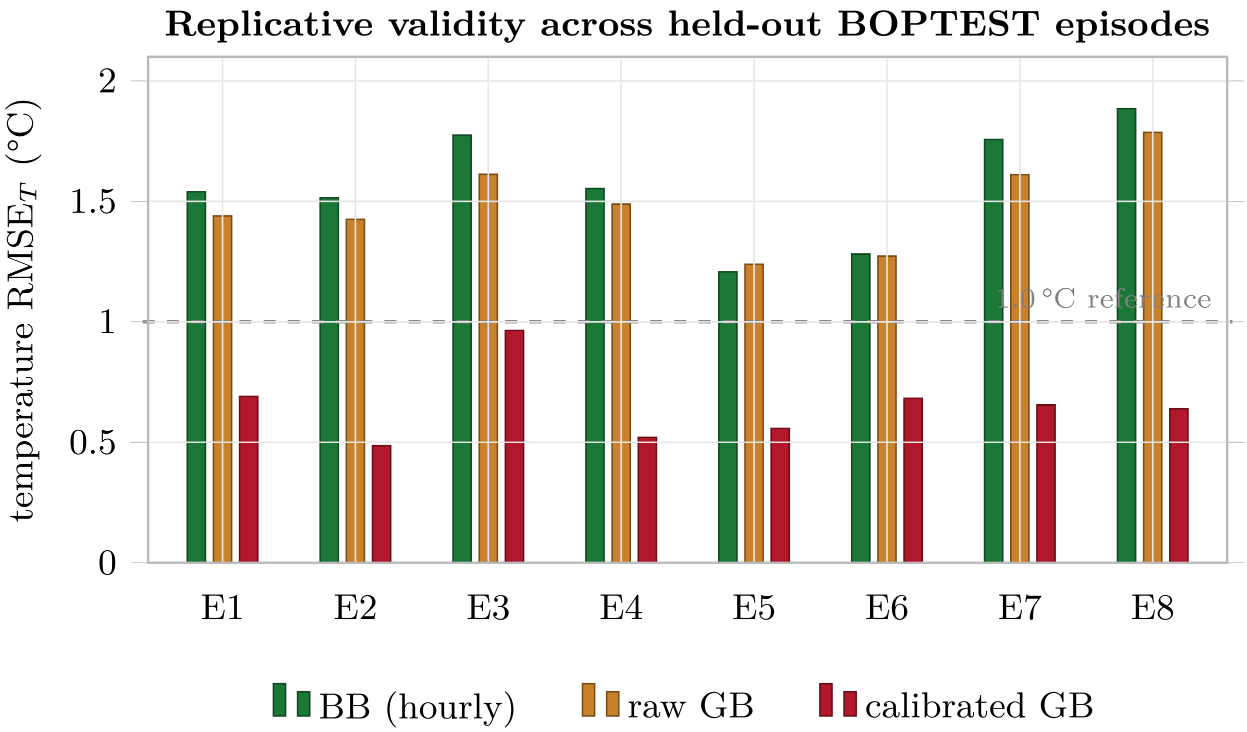


Figure S3: Replicative validity across the eight held-out BOPTEST episodes used by the predictive-validation prepared-rollout results. The dashed line marks a 1 °C engineering reference.

Table S2: Predictive-validation data corpora referenced by the experimental protocol.

Dataset	Rows	Step (s)	Policy mix	Scenario mix
BB canonical corpus	51,200	3600	cool, heat, mixed, random	autumn, spring, summer, winter
GB calibration corpus	10,744	900	hdrl, thermostatic	peak_heat_window, typical_heat_window
15-min exploration corpus	48,384	900	cool, heat, mixed, pulse, random, thermostatic_noise	autumn, spring, summer, winter

Table S3: Architecture and calibration summary for the two surrogate roles.

Surrogate	Role	Architecture / physics	Corpus	Training or calibration protocol
BB direct-TSup	Fast control-oriented rollout dynamics	8D dual-head MLP: HeatFlowNetV2 (7,105 params, LayerNorm/Tanh/residual), PowerNetV2 (1,377 params, Tanh/Softplus); total 8,482 params	51,200 hourly transitions	AdamW, lr=1e-3, batch=256, epochs=500
GB calibrated	Physics-informed predictive twin	RC-NeuralODE backbone with positive C_zon reparameterization and residual temperature/power heads	10,744 prepared 15-min transitions	Stage A latency/bias; Stage B 120 epochs; Stage C 60+80 epochs; C_zon = 4.413×10^5 J/K

Table S4: BB configuration from the active model class and Hou-and-Evins reporting results.

Item	Value
Input dimension	8
Temperature head	HeatFlowNetV2, 7,105 params
Power head	PowerNetV2, 1,377 params
Total parameters	8,482
Epoch budget	500
Batch size	256
Learning rate	1e-3
Optimizer	AdamW
Early stopping	patience=30; improvement_threshold=1e-6

Table S5: Feature scaling and physical constraints used by the surrogate interface.

Variable	Scaling method	Parameters	Engineering role
surrogate_t_zone	affine min-max to [-1, 1]	min=15 C; max=35 C	Keeps the learned thermal state within the training support used by the direct-TSup surrogate.
surrogate_t_amb	affine min-max to [-1, 1]	min=-10 C; max=40 C	Normalizes outdoor-air temperature for the heat and power heads.
hour	cyclic sine/cosine	sin(2*pi*hour/24); cos(2*pi*hour/24)	Avoids discontinuity between hour 23 and hour 0.
day	cyclic sine/cosine	sin(2*pi*day/365); cos(2*pi*day/365)	Represents seasonal phase without an artificial year-end jump.
a0_raw	native action domain	[-1, 1] maps to 18..35 C supply temperature	Preserves the PPO action coordinate while keeping physical supply-temperature bounds explicit.
a1_raw	native action domain	[-1, 1] maps to 0..1 fan command	Keeps fan modulation aligned with the controller action range.
power_head_output	Softplus multiplied by P_MAX	P_MAX=5500 W	Guarantees non-negative predicted HVAC power.

Table S6: Stage A telemetry-alignment operations from the calibration-summary JSON.

Operation	Estimated value	Effect
Latency compensation	1 step (15 min)	latency-search RMSE 0.378 C
Temperature bias removal	-0.1049 C	median residual offset
Power affine normalization	scale -0.702, bias 1148 W	global power-channel rescale
Post-Stage-A temperature RMSE	0.2383 C	baseline for Stage B
Post-Stage-A power MAE	514.4 W	baseline for Stage C

Table S7: Stage B excitation filtering and physical identification of C_{zon}.

Parameter	Value	Notes
C_zon prior	4.200×10^5 J/K	air volume × specific heat
C_zon after Stage B	4.413×10^5 J/K	+5.1% vs prior
Stage B epochs	120	episode-aware run
C_zon learning rate	10^{-3}	dedicated scalar LR
Excitation quantile	0.95	|dT| threshold 0.1759
Excitation rows	403 / 8058	mean score 0.240 vs 0.069

Table S8: Stage A/B/C calibration summary from the canonical JSON and rollout results.

Metric	Before	After	Relative change
1-step RMSE$_T$ (°C)	0.384	0.235	-38.8%
24 h rollout RMSE$_T$ (°C)	1.466	0.644	-56.1%
Power MAE (W)	808	482	-40.3%
C_{zon} ($\times 10^5$ J/K)	4.200	4.413	+5.1%

Table S9: Multi-horizon teacher-forced rollout fidelity from the Hou-and-Evins predictive-validity table. Cells are temperature RMSE (°C) at each horizon; the final column is the 24 h temperature R^2 (n/a where not computed for the auxiliary baselines).

Model	1 h	4 h	8 h	24 h	24 h R^2_T
BB (hourly)	1.471	1.579	1.579	1.557	-1.411
BB (15-min matched)	0.875	0.874	0.873	0.876	n/a
raw GB	1.491	1.491	1.491	1.466	n/a
calibrated GB	0.321	0.575	0.638	0.644	+0.588

Table S10: Backend throughput under the 900 s control protocol from the speed-benchmark result (single CPU thread, BOPTEST reached through the same HTTP-Docker interface used in training).

Backend	Steps/s	Median (ms)	P95 (ms)	Speed-up
BOPTEST RTE HTTP loop	21.0	14.915	19.251	1.0×
BB surrogate	4,626.3	0.165	0.273	220.2×
GB calibrated	2,399.5	0.359	0.495	114.2×
hybrid BB+GB	1,786.8	0.507	0.648	85.0×

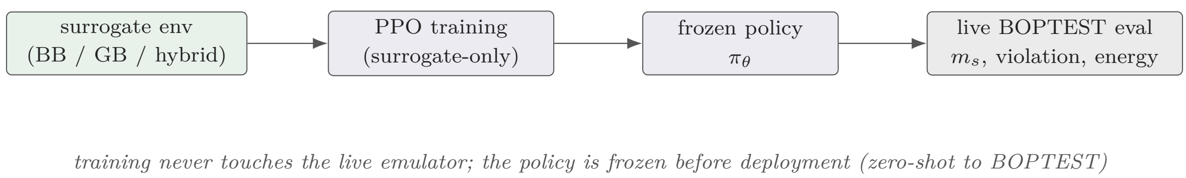


Figure S4: Controller-learning pipeline. The experiments separate the rollout model, the physical regularizer, the controller family, the observation interface, and live BOPTEST validation.

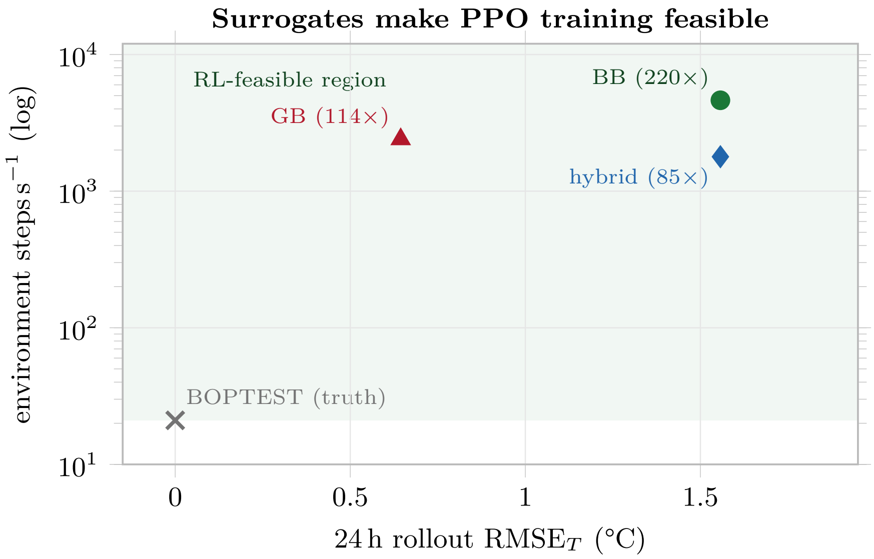
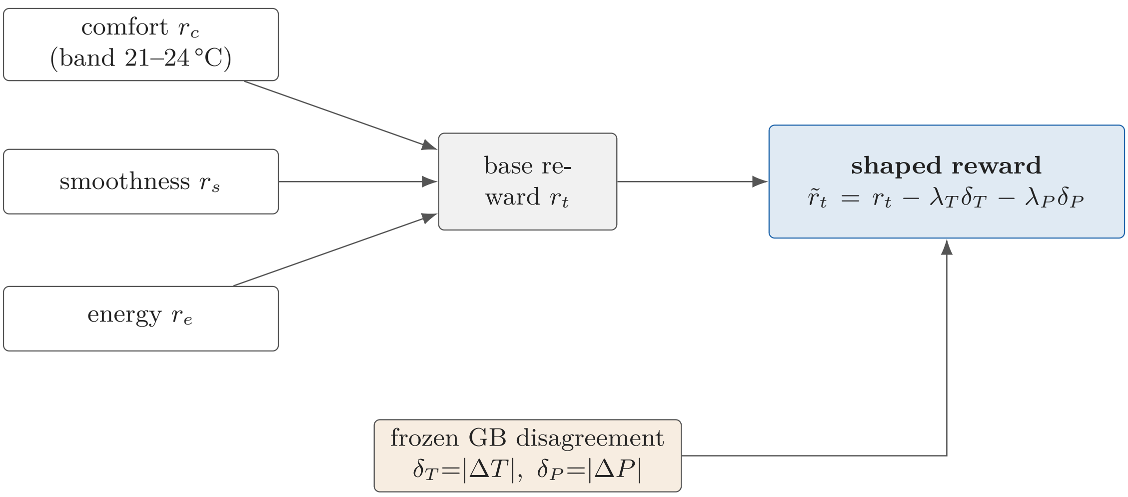


Figure S5: Surrogates make policy-gradient training computationally feasible. Environment throughput (env steps/s, log scale) versus 24 h rollout RMSE_T for the BOPTEST emulator (ground truth) and the three in-process surrogate backends. The surrogates run 85–220× faster than the BOPTEST RTE HTTP testbed, turning a days-long PPO run into hours; the hybrid is plotted at BB's rollout RMSE because it rolls out on BB. The plotted values combine measured throughput with 24 h rollout-RMSE estimates.



$\lambda_T = 0.10$, $\lambda_P = 5 \times 10^{-5}$; the GB penalty is forward-only and never enters the policy loss.

Figure S6: Hybrid reward-shaping mechanism. BB supplies rollout dynamics; frozen GB supplies a disagreement penalty on the same state-action transition.

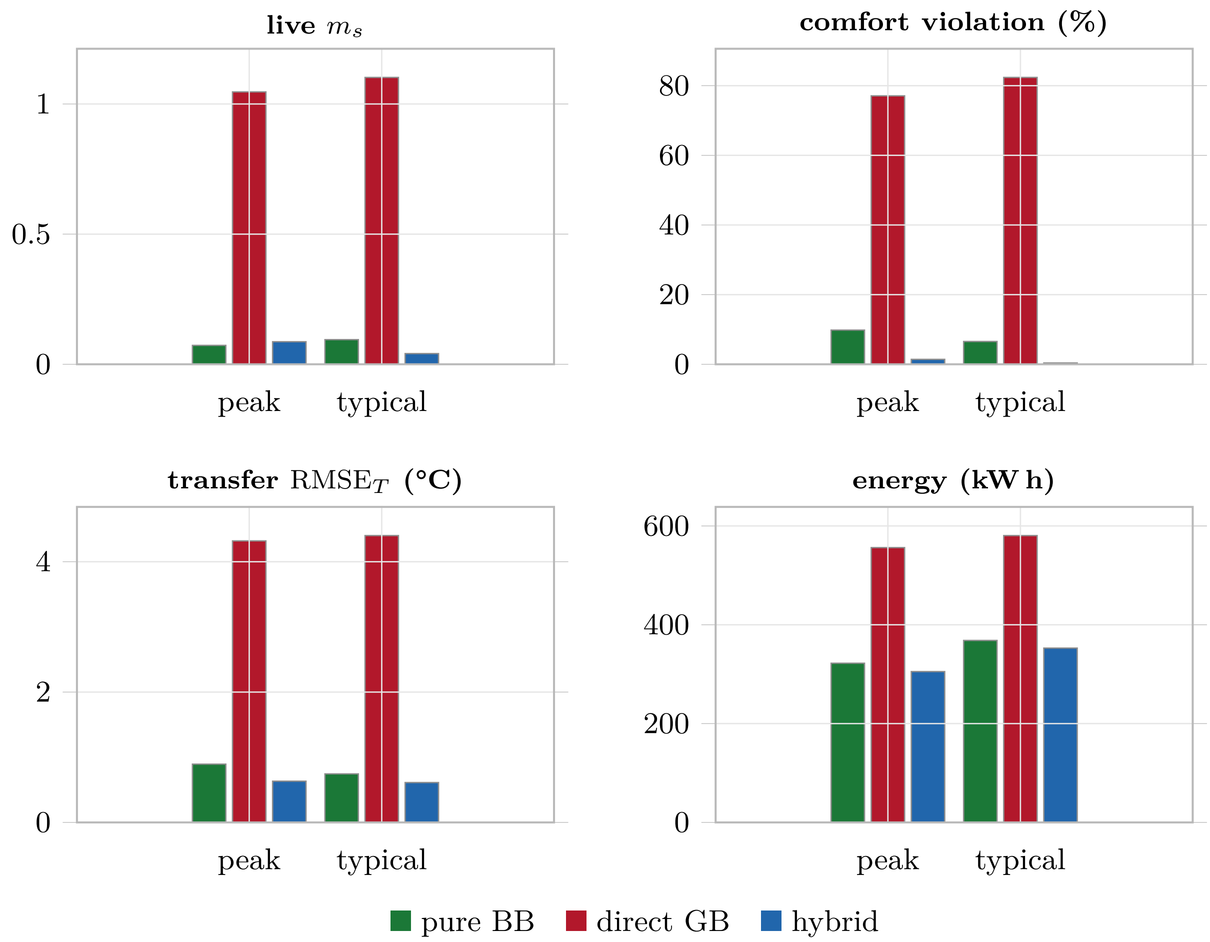


Figure S7: Detailed live BOPTEST KPI breakdown (per-metric bars) for pure BB, direct GB, and hybrid PPO. Direct GB is a negative control: higher predictive fidelity does not imply a useful RL training environment. These per-metric bars give the underlying m_s, violation, RMSE, and energy behind the compact fidelity–utility scatter in the main text.

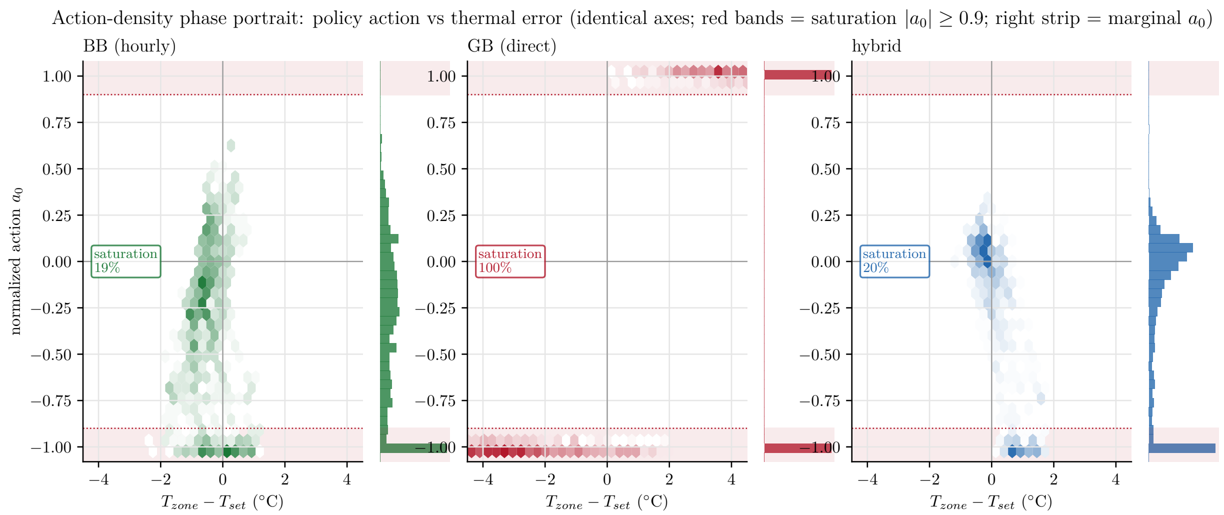


Figure S8: Action-density phase portrait of the three thermostatic controllers on the live BOPTEST `typical_heat_window`: empirical hexbin density of the normalized supply-temperature action a_0 (one hue per backend) against the thermal error $T_{\text{zone}} - T_{\text{set}}$, on *identical* axes, with the marginal a_0 distribution in the right strip of each panel and the measured saturation share ($|a_0| \geq 0.9$). The red bands mark the actuator-saturation region. Direct GB collapses onto $a_0 = -1$ (100% saturation, a bang-bang law) whereas the usable hourly BB and the hybrid modulate (≈19–20%).

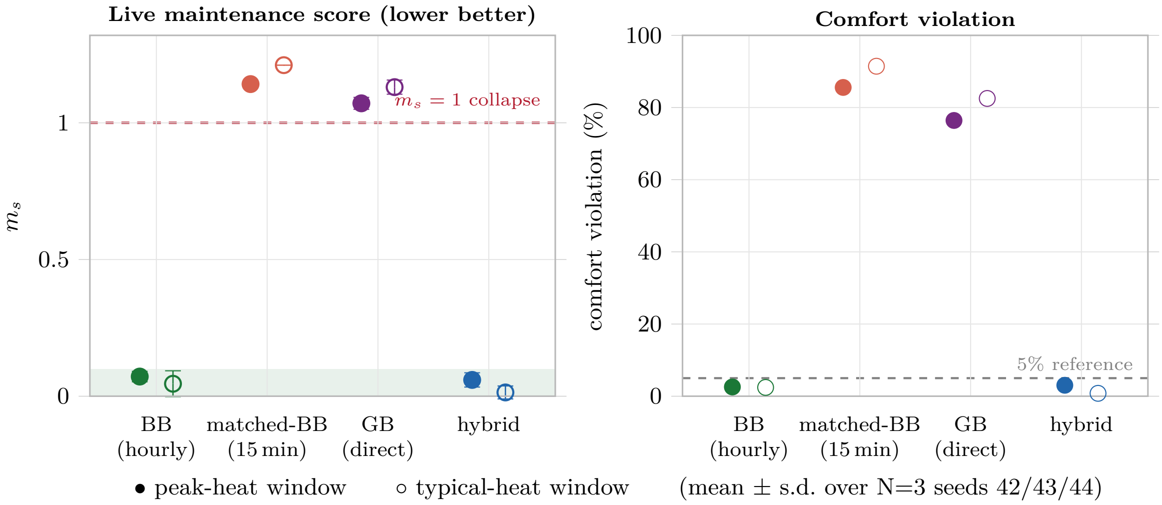


Figure S9: Across $N = 3$ seeds the fidelity–utility paradox is not a seed result: the usable hourly-BB and hybrid controllers stay below both the $m_s = 1$ closed-loop-collapse line and the 5% comfort-violation reference on every seed, while the matched-resolution BB and the direct GB both collapse on every seed ($m_s > 1$, $> 75\%$ violation) with small seed variance. Markers are the mean over seeds {42,43,44} (filled = peak-heat window, open = typical-heat window); whiskers are ± one standard deviation; colour encodes the controller (green = coarse BB, salmon = matched-resolution BB, purple = direct GB, blue = hybrid). **(left)** live maintenance score m_s (lower is better, with the usable $m_s < 0.1$ band shaded); **(right)** comfort-violation fraction. Same data as Table S15.

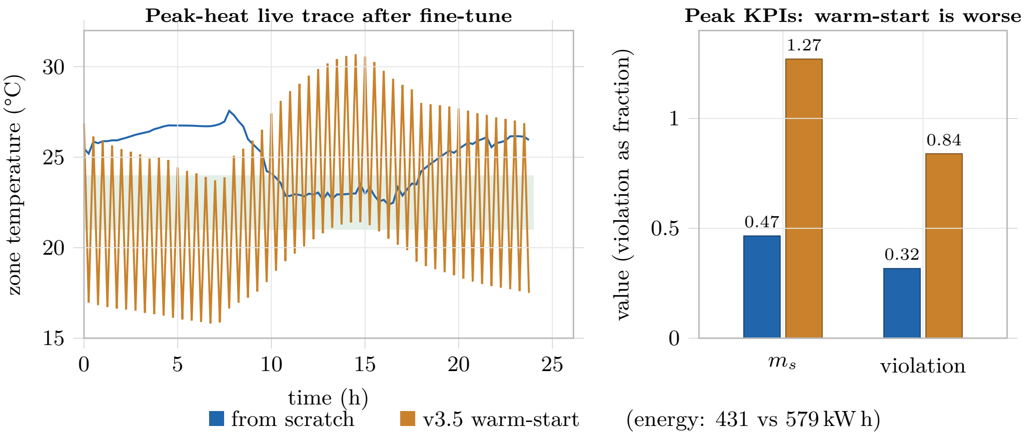


Figure S10: Warm-start negative control. Pretraining on direct GB and then fine-tuning on the hybrid backend is inferior to training the hybrid policy from scratch.

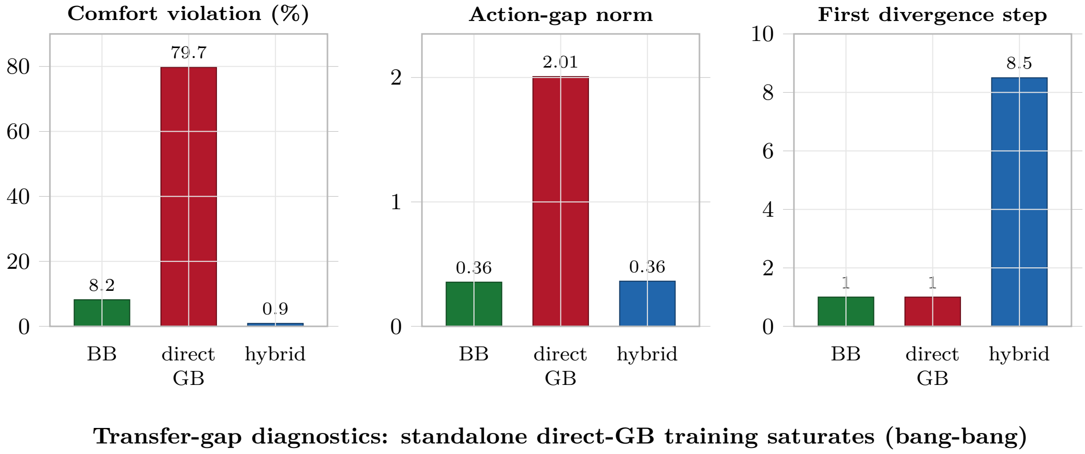


Figure S11: Transfer-gap diagnostics. Direct GB has high action-gap norm and live-surrogate mismatch; the hybrid backend suppresses both.

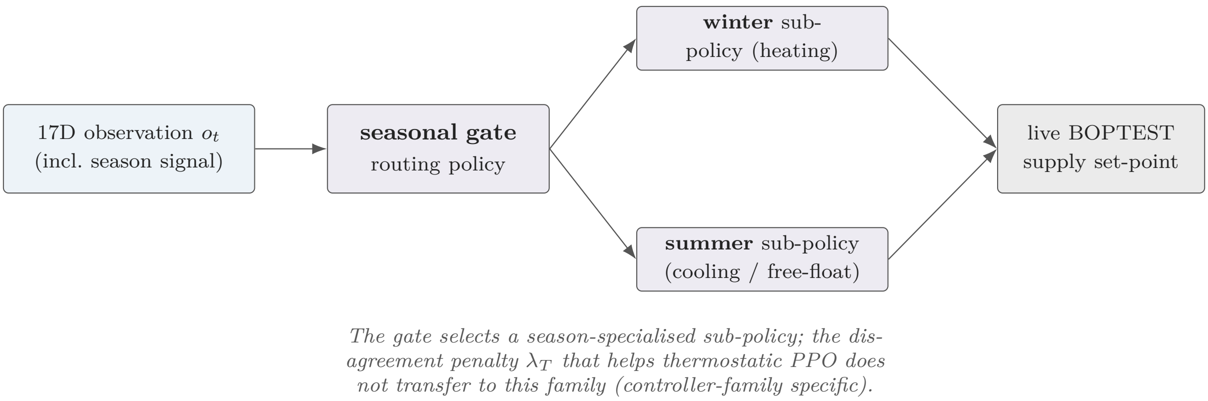


Figure S12: HDRL architecture: a high-level seasonal gate $k(t)$ routes the 17D observation to one of two low-level PPO setpoint specialists (winter / summer). Because the low level is already comfort-aware, the GB temperature censor over-regularizes it.

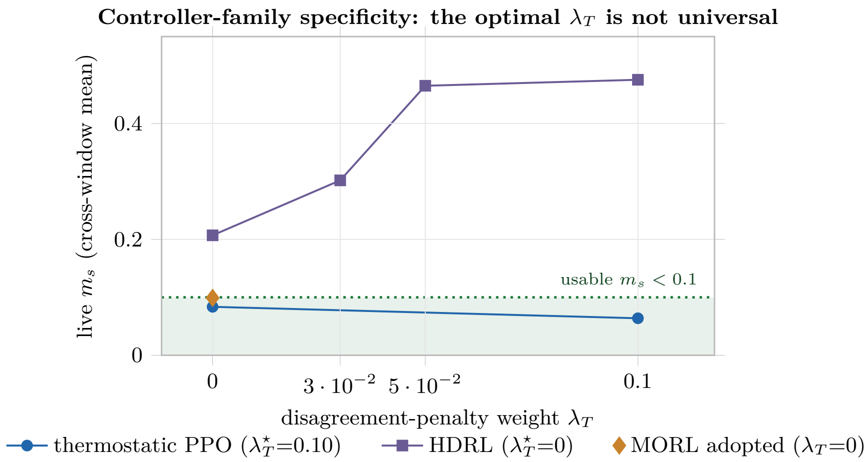


Figure S13: The optimal censor weight λ_T is controller-family specific. Live BOPTEST maintenance score m_s (cross-window mean of the peak and typical windows) versus λ_T for each controller family; $\star$ marks the per-family optimum. Thermostatic PPO improves with the censor (optimum $\lambda_T = 0.10$, the canonical hybrid), HDRL is best *without* it ($\lambda_T = 0.00$) and degrades monotonically, and MORL adopts $\lambda_T = 0.00$. HDRL has a dense four-point sweep; PPO is shown at its two measured endpoints (pure BB = $\lambda_T 0$, hybrid = 0.10) and MORL at its single adopted configuration. The panel compares the *location* of each optimum, not absolute m_s across families (they use different observation/action interfaces).

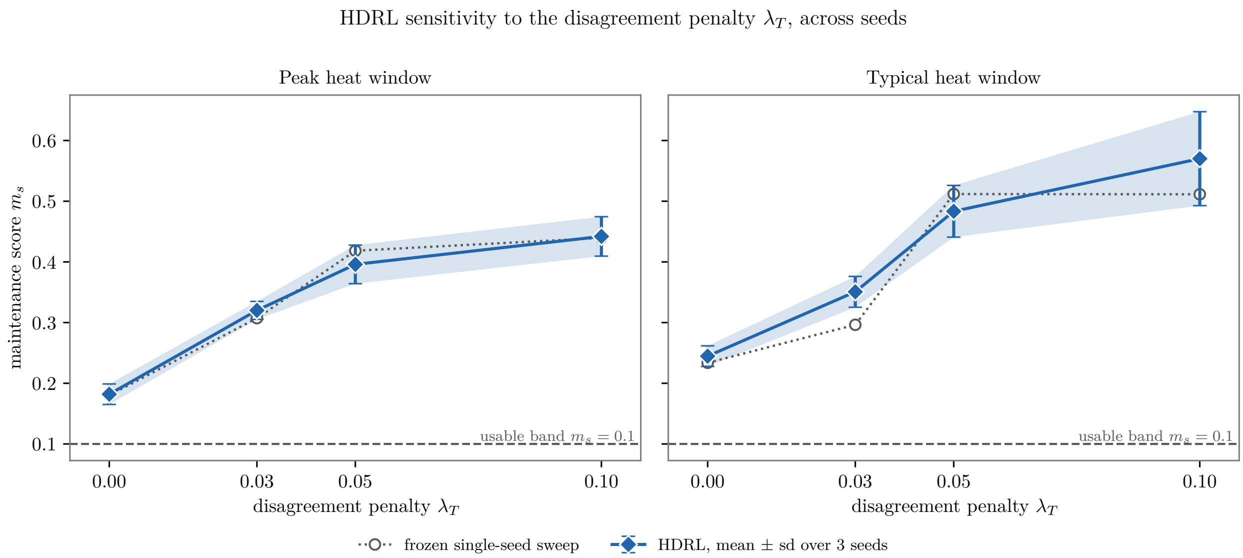


Figure S14: HDRL λ_T sensitivity, mean ± s.d. over seeds $\{42, 43, 44\}$ at each swept weight; the frozen single-seed sweep is overlaid as hollow markers. The thermostatic regularization weight does not transfer: the best HDRL policy uses no temperature-disagreement term, and the λ_T=0 → 0.10 endpoint gap is 10.0 (peak) and 5.8 (typical) pooled seed standard deviations.

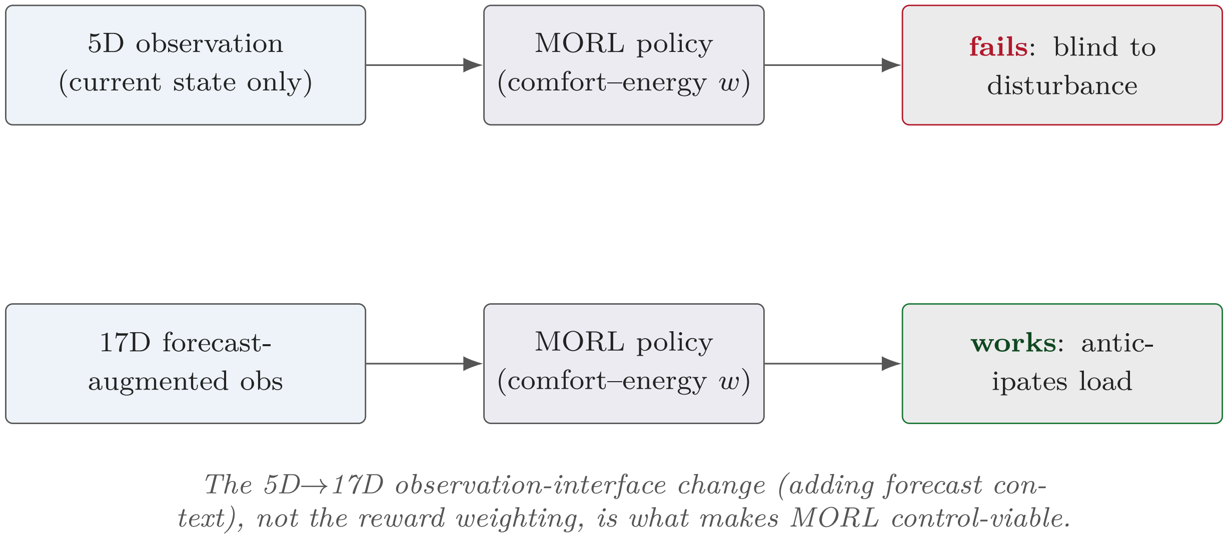


Figure S15: MORL observation-interface schematic. The 5D instantaneous observation fails because it is blind to forecasted disturbances, whereas the 17D forecast-augmented observation makes MORL control-viable. The quantitative ablation of this schematic is in Figure S16.

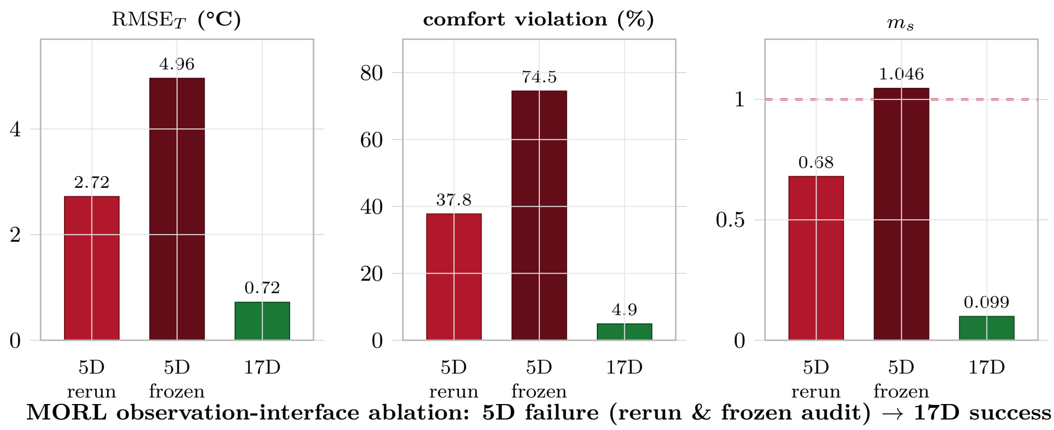


Figure S16: MORL observation-interface ablation. The main article reports the headline 5D-versus-17D numbers in its controller-family boundary-test paragraph; this supplementary plot includes the current-code 5D rerun, the frozen historical 5D audit result, and the 17D power-only recovery.

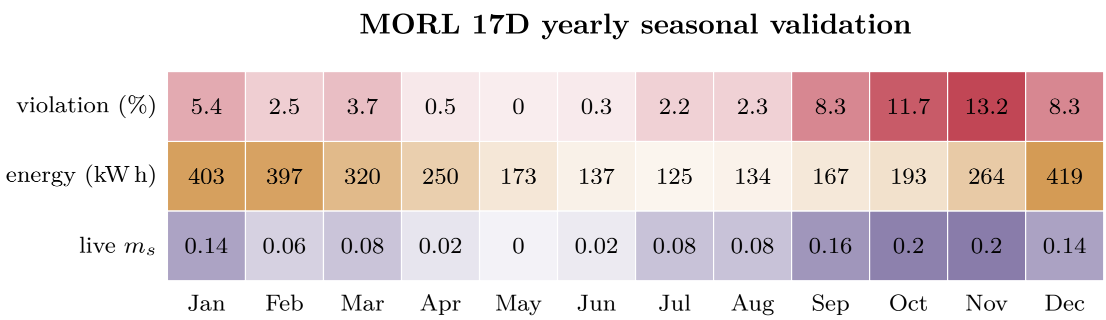


Figure S17: MORL seasonal performance heatmap under the 17D interface. The figure shows monthly structure but does not support the N=3 mechanism claim after N=5 extension.

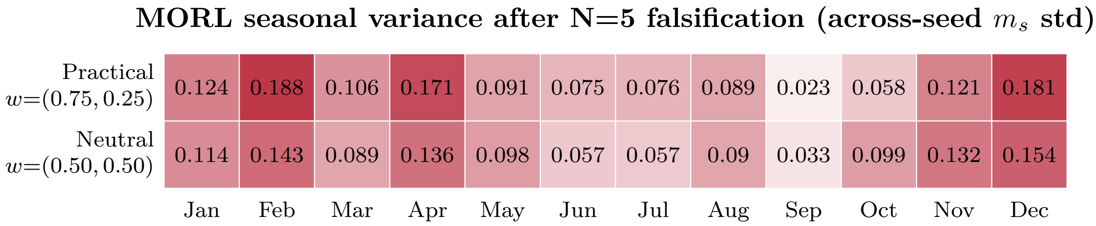


Figure S18: Post-N=5 seasonal variance diagnostic. The earlier seasonal-inversion hypothesis is reported as falsified, not retained as a positive mechanism claim.

Table S11: Targeted-window and yearly scenario definitions.

Scenario	Day idx	Start (s)	Dur. (d)	Ambient (°C)	Role
`peak_heat_window`	3	259200	14	-24.4	January coldest, heating stress test
`typical_heat_window`	37	3196800	14	2.4	February moderate, deployment-realistic
yearly evaluation	12 months	—	14/mo	varied	MORL + PI yearly summary

Table S12: PPO training configuration used in the controller-learning experiments.

Parameter	Thermostatic PPO	HDRL	MORL pretrain	MORL finetune
Algorithm	PPO, MlpPolicy	PPO, MlpPolicy	PPO, MlpPolicy	PPO load+continue
Learning rate	3.0×10^{-4}	3.0×10^{-4}	3.0×10^{-4}	1.0×10^{-4}
`n_steps`	1024	1024	2048	inherited
`batch_size`	4096	2048	64	inherited
`n_epochs`	10	10	10	10
γ	0.99	0.99	0.99	0.99
GAE λ	0.95	0.95	0.95	0.95
Clip range	0.20	0.20	0.20	0.20
Total timesteps	10M	12M	2M	100k
Seed	42	42	42	42–46 (N=5)

Table S13: Extended 17D observation feature groups.

Feature group	Dim.	Contents
Physical state (basic)	5	T_{zone}, CO_2, clipped-log power, prev. T_{sup}, T_{amb}
Cyclic time	4	hour and day sine/cosine encodings
Ambient forecast	5	T_{amb} at $+1, +3, +6, +12, +24$ h
Previous action	2	$(a_{T_{\text{sup}}}, a_{\text{fan}})$ from last step
Temperature delta	1	causal-smoothed ΔT_{zone}
Total (extended)	17	obs_mode = extended

The shaped training reward is $r_t = -(w_c\, c_t + w_e\, e_t)$, with a piecewise-linear, asymmetric comfort term

$$r_{\text{comfort},t} = \begin{cases} -w_u\,(T_\ell - T_t) + b_{\text{heat}} & T_t < T_\ell, \\ +\beta & T_\ell + \delta \le T_t \le T_h - \delta, \\ -w_o\,(T_t - T_h) + b_{\text{cool}} & T_t > T_h, \end{cases} \tag{1}$$

and an energy term $r_{\text{energy},t} = -\eta P_t$ ($\eta = 2 \times 10^{-4}$); the cold/hot weights are ambient-dependent and fixed a priori. The parameters are listed below.

Table S14: Reward-shaping parameters.

Component	Value	Role
Comfort band	21.0 / 24.0 C	comfort interval (band_low/high)
Comfort deadband	0.50 C	soft margin around band edges
In-band bonus	0.050/step	reward for staying inside the band
Undershoot / overshoot weight	1.15 / 1.15	generic violation penalties
Cold-ambient asymmetry	1.60 (amb < 8.0 C)	extra cold-undershoot penalty
Hot-ambient asymmetry	1.80 (amb > 24.0 C)	extra hot-overshoot penalty
Heating action bonus	0.04 ($T_{\text{sup}} \geq 29.0$ C)	anti-degenerate shaping
Cooling action bonus	0.06 ($T_{\text{sup}} \leq 21.0$ C)	anti-degenerate shaping
MORL weights ($w_c/w_e/w_s$)	0.80 / 0.20 / 0.00	canonical scalarization
Energy scale	$2e-4$ (W→reward)	energy-to-reward conversion

Table S15: Seed robustness ($N = 3$ seeds {42,43,44}) of the thermostatic controllers on the live BOPTEST targeted windows; mean ± standard deviation. The usable controllers (pure BB, hybrid) stay below the 5% comfort-violation bar on every seed, while the collapse controls (matched-resolution BB, direct GB, and the Δt-rescaled hourly BB) exceed $m_s = 1$ on every seed with small seed variance. The Δt-rescaled control keeps the hourly BB but rescales its increment to the 900 s runtime step; its collapse shows the hourly BB's usability rests on its large coarse-scale increment.

Controller	Window	N	m_s (mean ± std)	Violation % (mean ± std)
pure BB	peak	3	0.072 ± 0.019	2.6 ± 1.2
pure BB	typical	3	0.045 ± 0.047	2.4 ± 2.2
matched BB (15-min)	peak	3	1.142 ± 0.000	85.6 ± 0.0
matched BB (15-min)	typical	3	1.211 ± 0.000	91.4 ± 0.0
direct GB	peak	3	1.071 ± 0.022	76.4 ± 0.7
direct GB	typical	3	1.131 ± 0.026	82.5 ± 0.7
Δt-rescaled BB	peak	3	1.187 ± 0.010	83.2 ± 0.7
Δt-rescaled BB	typical	3	1.215 ± 0.006	86.0 ± 0.3
hybrid (λ_T=0.10)	peak	3	0.059 ± 0.026	3.0 ± 1.6
hybrid (λ_T=0.10)	typical	3	0.014 ± 0.024	0.8 ± 1.4

Table S16: Measured response-surface smoothness of the three single-model backends, probed directly on each surrogate's one-step action map (no controller, no BOPTEST). The scale-free relative roughness (curvature normalized by slope) is independent of the step length and tracks closed-loop usability: the usable hourly BB is the smoothest; both collapsed backends are several times rougher.

Surrogate backend	Slope $\overline{\lvert\partial\hat{T}/\partial a_0\rvert}$	Rel. roughness	vs. BB	Closed-loop role
canonical BB (hourly)	3.368	0.169	1.0 (ref)	usable
matched BB (15-min)	0.272	1.578	9.4×	collapse
calibrated GB	0.467	1.338	7.9×	collapse

Table S17: Thermostatic hybrid λ_T sweep *design* (experimental protocol §5; $\lambda_P = 5 \times 10^{-5}$ throughout); the mid point is the retained canonical. Of these pre-specified brackets, the live-BOPTEST two-window evaluation reported in the paper covers thermostatic PPO at $\lambda_T = 0.10$ (canonical hybrid) and $\lambda_T = 0$ (pure BB), so the controller-family λ_T figure shows PPO at these two measured endpoints; the 0.05 and 0.15 brackets were not carried to full live two-window evaluation. The dense four-point λ_T sweep is the HDRL family, not PPO.

λ_T	Tag	Role
0.05	hybrid_l005	weaker censor (sweep bracket)
0.10	hybrid_l010	**canonical** (retained; $m_s = 0.087$ peak, 0.041 typical)
0.15	hybrid_l015	stronger censor (sweep bracket)

Table S18: Direct-GB warm-start utility.

Mode	Scenario	m_s	Violation (%)
scratch (random init)	peak	0.465	31.70
scratch (random init)	typical	0.578	45.54
warm-start (from GB)	peak	1.270	83.93
warm-start (from GB)	typical	1.289	86.83

Table S19: Transfer diagnostics across three backends. Δm_s = surrogate − live (negative ⇒ surrogate optimistic); Temp gap is the live closed-loop RMSE_T.

Variant	Scenario	Temp gap (°C)	Δm_s	Action gap	First div.	Top driver
pure BB	peak	0.894	-0.102	0.377	1	p_total_norm
pure BB	typical	0.745	-0.068	0.333	1	p_total_norm
hybrid $\lambda_T = 0.10$	peak	0.633	-0.023	0.473	1	p_total_norm
hybrid $\lambda_T = 0.10$	typical	0.612	-0.021	0.253	16	p_total_norm
direct GB	peak	4.320	-0.886	2.000	1	t_zone_norm
direct GB	typical	4.401	-1.014	2.014	1	t_zone_norm

Table S20: HDRL sweep over λ_T on the targeted windows, mean ± s.d. over seeds $\{42, 43, 44\}$. Best is $\lambda_T = 0$; m_s increases monotonically in the seed means as the temperature censor strengthens, though the $0.05 \rightarrow 0.10$ increment lies within one pooled seed s.d. on both windows.

λ_T	Scenario	m_s	Violation (%)	RMSE_T (°C)	Energy (kWh)
0.00	peak	0.182 ± 0.017	6.27 ± 1.69	0.819 ± 0.059	311.8 ± 4.8
0.03	peak	0.320 ± 0.015	12.00 ± 0.37	1.023 ± 0.030	304.4 ± 2.1
0.05	peak	0.396 ± 0.032	19.77 ± 3.08	1.245 ± 0.083	301.1 ± 2.4
0.10	peak	0.442 ± 0.033	24.43 ± 3.83	1.323 ± 0.031	303.7 ± 8.9
0.00	typical	0.245 ± 0.017	4.09 ± 1.83	0.822 ± 0.044	368.0 ± 4.9
0.03	typical	0.351 ± 0.025	14.43 ± 3.31	1.139 ± 0.024	362.1 ± 4.0
0.05	typical	0.483 ± 0.043	27.65 ± 4.28	1.392 ± 0.092	353.7 ± 3.2
0.10	typical	0.570 ± 0.077	34.52 ± 6.42	1.501 ± 0.118	359.3 ± 8.1

Table S21: MORL Pareto (seed 42) and N=5 canonical seed analysis.

Preference / statistic	RMSE_T (°C)	Violation (%)	m_s	Interpretation
0/100 (seed 42)	8.359	86.76	1.588	energy-only collapse
25/75 (seed 42)	0.912	11.39	0.173	energy-weighted usable
50/50 (N=5 mean±std)	0.893±0.081	13.01±6.62	0.187±0.078	neutral, CV=0.42
75/25 (N=5 mean±std)	0.799±0.100	9.23±6.67	0.139±0.085	practical, CV=0.61
100/0 (seed 42)	0.631	1.49	0.032	best seed-42 comfort

Table S22: Per-seed MORL yearly metrics for the two canonical weight pairs.

Pair (c/e)	Seed	RMSE_T	Within 1°C (%)	Violation (%)	Energy (kWh)	m_s
50/50	42	0.939	74.9	12.92	2843.6	0.193
50/50	43	0.769	82.9	6.77	2872.0	0.103
50/50	44	0.879	76.9	9.34	2858.4	0.141
50/50	45	0.895	77.7	11.97	2839.8	0.187
50/50	46	0.985	65.4	24.05	2554.1	0.310
75/25	42	0.700	83.3	3.08	2994.3	0.057
75/25	43	0.716	84.9	5.47	2963.9	0.087
75/25	44	0.769	81.7	7.82	2819.1	0.117
75/25	45	0.903	78.5	9.42	3036.7	0.160
75/25	46	0.909	70.3	20.37	2602.8	0.276

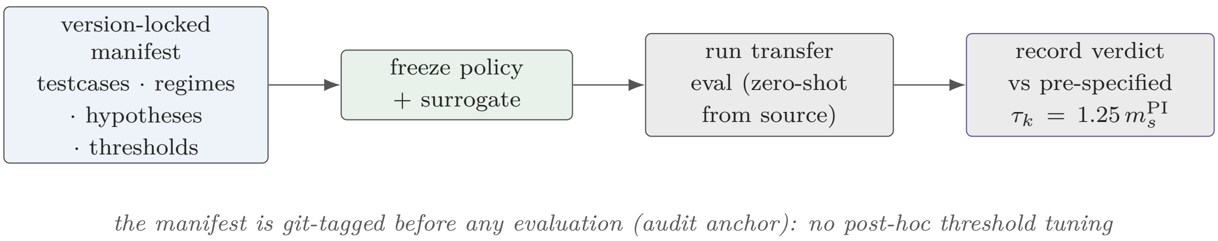


Figure S19: Pre-specified transferability protocol. Controller-side transfer is tested against a testcase-specific PI threshold; surrogate-side transfer is tested by full Stage A/B/C recalibration on target telemetry.

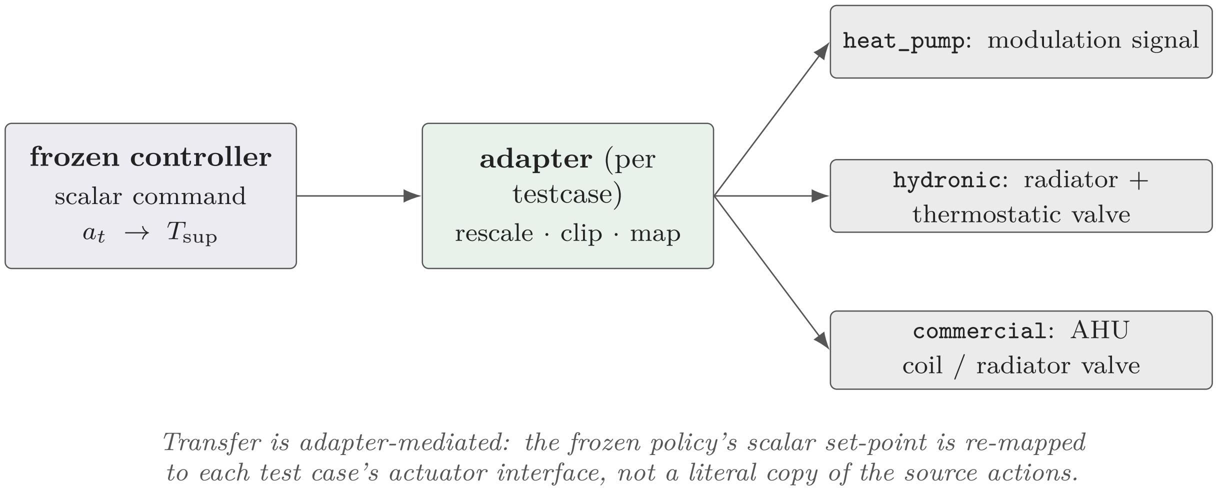


Figure S20: Adapter-mediated transfer schematic. The frozen hybrid policy is routed through a pre-specified actuator adapter $\mathcal{A}_k$ to each testcase's actuator interface; the controller-side verdict (against the 1.25× PI threshold) is per-testcase, while full Stage A/B/C surrogate recalibration passes on all three.

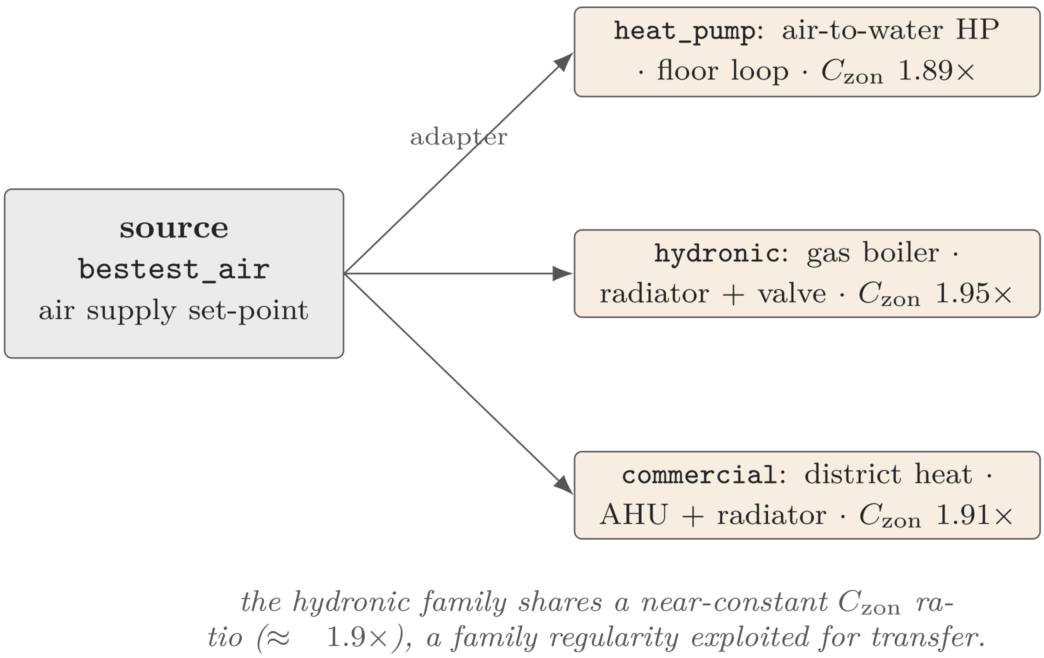


Figure S21: Schematic (no data). Comparative HVAC topology of the source case and the three hydronic targets (source → distribution → zone). Transfer changes the heat source, distribution path, and actuator set, while the single-zone energy balance and the zone thermal capacitance C_{zon} are shared across the family.

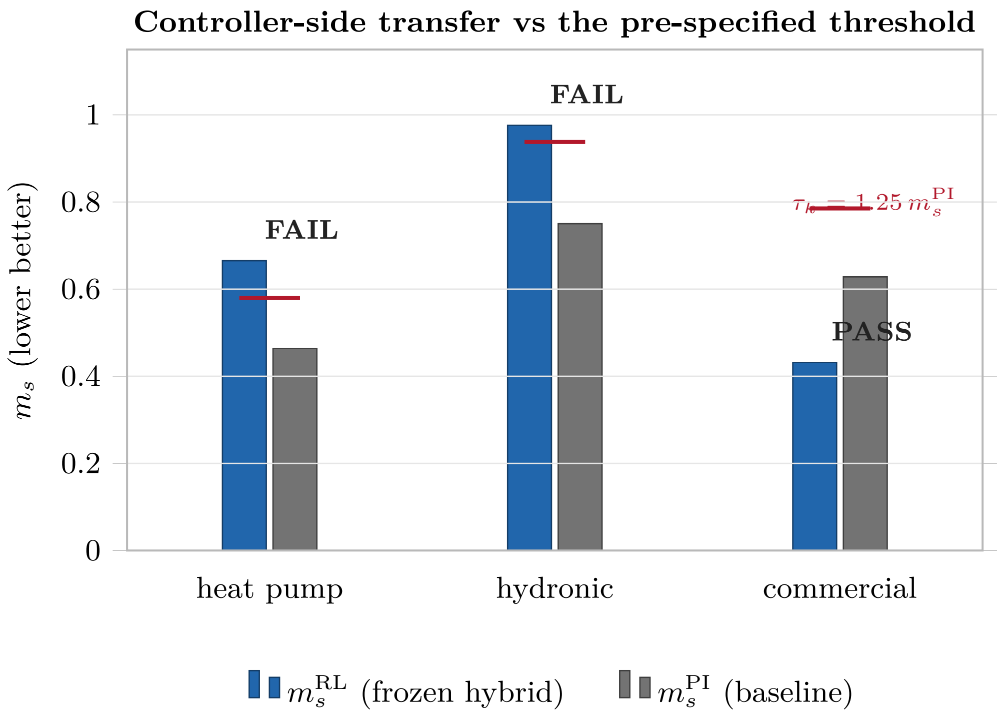


Figure S22: Controller-side transfer: frozen-hybrid $m_{s,\mathrm{RL}}$ against the PI baseline $m_{s,\mathrm{PI}}$ and the pre-specified threshold $\tau = 1.25\, m_{s,\mathrm{PI}}$ (dashed). The two residential cases exceed the threshold (FAIL); the commercial case is below it (PASS, with the separate energy caveat).

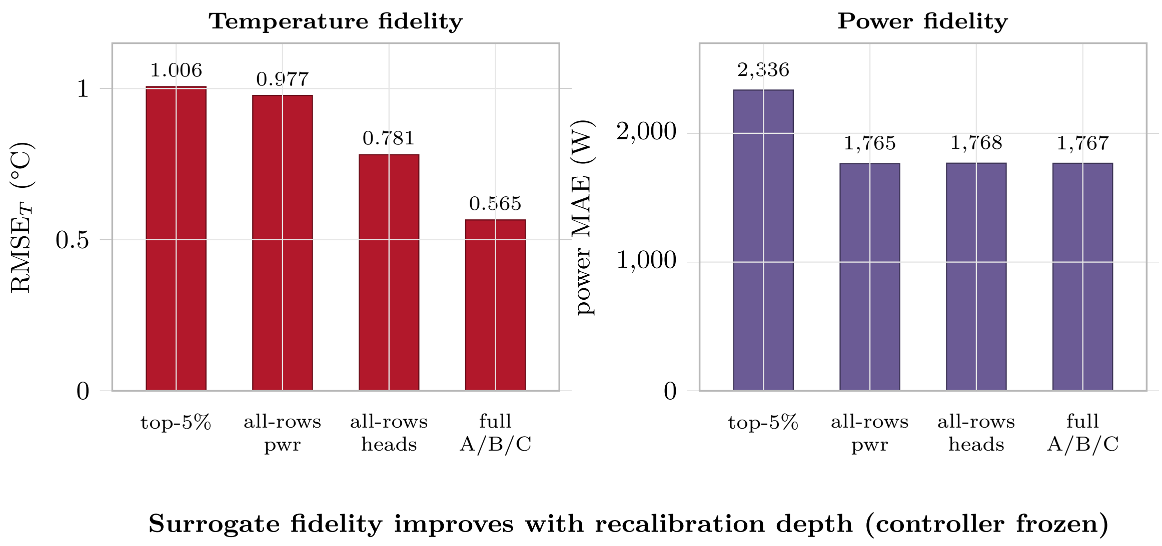


Figure S23: Primary-testcase surrogate-fidelity progression across recalibration regimes. Temperature and power fidelity improve monotonically toward full Stage A/B/C, while the frozen controller's live m_s is invariant by experimental design.

Hypothesis closure vs version-locked predictions

Hypothesis / prediction	Observed closure	Evidence
H1 strong	FALSIFIED	frozen recipe, direct transfer
H2 medium	FALSIFIED (by structural definition)	partial recalibration
H3 surrogate	SUPPORTED (on N=3)	full Stage A/B/C RMSE
H3 controller	FALSIFIED (regime-dependent failure)	controller full regime
Stretch none	expected FAIL → observed PASS	commercial, mode=none
Stretch C_{zon}	expected scale-dep. → observed 1.909×	commercial C_{zon}
Stretch RMSE	expected 50–90% → observed 87.81%	commercial recalibration

Figure S24: Hypothesis closure matrix: the transfer-boundary experiment supports surrogate-side transferability but falsifies deployment-ready frozen-controller transfer.

Table S23: Pre-specified target testcases and actuator adapters.

Role	Testcase	Structural difference vs `bestest_air`	Adapter
Primary	`bestest_hydronic_heat_pump`	hydronic loop driven by a heat pump	`hydronic_t_supply_v1`
Secondary	`bestest_hydronic`	boiler/radiator hydronic distribution	`hydronic_direct_supply_v1`
Stretch	`singlezone_commercial_hydronic`	order-of-magnitude larger commercial zone + hydronic valve	`commercial_hydronic_valve_v1`

Table S24: Per-testcase actuator-adapter mapping. All adapters share $h = \text{clip}((T_{\text{sup},t} - 18)/17, 0, 1)$; setpoint overrides are in kelvin.

Testcase	Primary override	Override formula	Auxiliary overrides
`heat_pump`	`oveTSet_u` (K)	$288.15 + 9h$ (zone setpoint 15–24°C)	`oveHeaPumY_u`, `ovePum_u`, `oveFan_u` from h
`hydronic`	`oveTSetSup_u` (K)	$T_{\text{sup},t} + 273.15$ (direct supply)	`ovePum_u` from h; fixed comfort-band context
`commercial`	`dh_oveTSupSetHea_u` (K)	$T_{\text{sup},t} + 273.15$ (district supply)	heating valves + pump from h; fixed zone/air context

Table S25: Pre-specified recalibration regimes. The controller is frozen in every regime.

Regime	Surrogate action	Controller action
none	frozen `bestest_air` surrogate recipe	frozen hybrid controller
partial	Stage C residual-head recalibration only	frozen hybrid controller
full	Stage A + Stage B + Stage C on target telemetry	frozen hybrid controller

Table S26: Testcase architecture and identified physics. C_{zon} in 10^5 J/K; equivalent mass = $C_{\text{zon}}/c_{p,\text{air}}$; mean delivered power from the Stage-C telemetry.

Testcase	Heat source	C_{zon}	eq. mass (kg)	mean power (W)	Thermal regime
`bestest_air` (source)	air supply	4.413	439	—	reference
hydronic heat pump	heat pump	8.348	831	2,778	conduction-dominated
hydronic (boiler)	boiler/radiator	8.623	858	679	slow hydronic
commercial hydronic	district + AHU	8.426	838	76,860	ventilation/AHU-dominated

Table S27: Primary-testcase per-regime detail. m_s is invariant across rows because the controller is frozen by experimental design; only surrogate fidelity changes.

Regime / result	RMSE_T (°C)	Power MAE (W)	$m_{s,\text{RL}}$	Status
PI baseline (yearly)	–	–	0.464	reference
none: frozen controller	–	–	0.665	control FAIL
partial: Stage C top-5%	1.006	2336	0.665	diagnostic
partial: all-rows power head	0.977	1765	0.665	diagnostic (power)
partial: all-rows heads	0.781	1768	0.665	surrogate cond. pass
full: Stage A/B/C	0.565	1767	0.665	surrogate PASS, ctrl FAIL

Table S28: C_{zon} re-identification across the `bestest_air` baseline and the three hydronic testcases.

Testcase	C_{zon} (J/K)	ρ_C vs `bestest_air`
`bestest_air` (source case)	4.413×10^5	1.000 (baseline)
hydronic heat pump	8.348×10^5	1.892
hydronic (boiler)	8.623×10^5	1.954
commercial hydronic	8.426×10^5	1.909
hydronic-family mean ± std	—	1.918 ± 0.032

Table S29: Stretch-testcase pre-specified predictions versus observed outcomes.

Prediction	Pre-reg.	A-priori	Observed	Verdict
mode=none controller verdict	FAIL	0.80	threshold PASS	FALSIFIED
full surrogate RMSE gain	$[50, 90]$%	0.70	87.8%	CONFIRMED
C_{zon} hyp. A (uniform)	$[1.7, 2.2]\times$	0.35	1.91×	CONFIRMED
C_{zon} hyp. B (scale-dep.)	$[3, 10]\times$	0.50	1.91×	FALSIFIED
C_{zon} hyp. C (failure)	non-convergence	0.15	converged (N=3)	FALSIFIED

5 BOPTEST RTE control loop and supply-temperature actuation

The live evaluation runtime is the BOPTEST Run-Time Environment (RTE): a Dockerised Modelica emulator with embedded baseline local-loop and supervisory controllers, a disturbance forecaster, and a KPI calculator, all exposed through a RESTful HTTP API (Blum et al., 2021). The test controller does not replace the plant; following the BOPTEST formalism it *overwrites* a chosen subset u_c of the emulator's control inputs while the remaining embedded loops keep running at baseline. In this study u_c is the single supply-temperature set-point of each test case, activated through the BOPTEST overwrite pair (the point's `_activate` flag set to 1 and its `_u` value written); on `bestest_air` this point is the supply-air-temperature set-point `oveTSetSup`. At every control step $\Delta t = 900\,\mathrm{s}$ (Option-1 synchronisation, in which the controller defines the step and the emulator advances when an action is returned) the controller reads measurements y_t (zone temperature, CO_2, power) and disturbance forecasts ω_t from the API, maps its policy output $a_t \in [-1, 1]$ to the supply-temperature command $T_{\mathrm{sup}} = 18 + \frac{a_t+1}{2}(35 - 18)\,^{\circ}\mathrm{C}$, writes it to the overwrite signal, and issues `POST /advance` to integrate the emulator forward one step. The RTE then returns the next measurements and the live key performance indicators (including the maintenance score m_s) computed from the closed-loop trajectory rather than from the surrogate. For the hydronic transfer cases the same supply-temperature command is routed through the per-test-case adapter to that case's overwrite signal (heat-pump modulation, valve position, or supply-water set-point), so the frozen policy interface is unchanged across test cases (Figure S25).

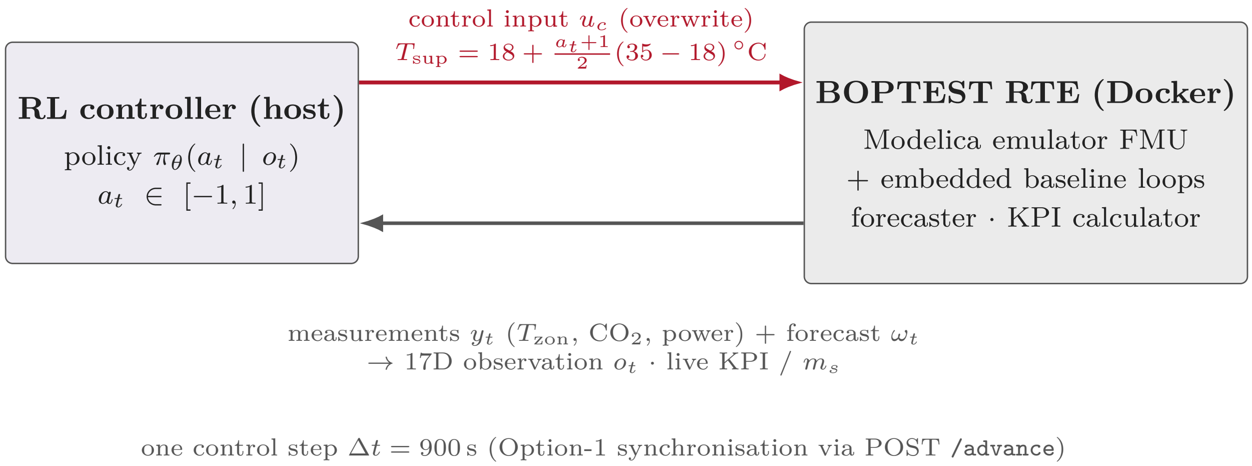


Figure S25: BOPTEST RTE closed-loop control interface (topology after Blum et al., 2021, Figs. 1–3). At each 900 s step the policy overwrites only the supply-temperature set-point through the `oveTSetSup` signal pair and advances the Dockerised emulator, which returns measurements, forecasts, and the live maintenance score m_s.

6 Detailed formulas

The following standard or implementation-level equations were moved here to keep the main article focused on results; they are reproduced for completeness. **PPO clipped surrogate objective** (Schulman et al., 2017), with $\hat{A}_t$ the GAE advantage:

$$L_{\mathrm{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\big(\rho_t(\theta)\,\hat{A}_t,\ \mathrm{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon)\,\hat{A}_t\big)\Big], \qquad \rho_t(\theta) = \frac{\pi_\theta(a_t \mid o_t)}{\pi_{\theta_{\mathrm{old}}}(a_t \mid o_t)}. \tag{2}$$

Black-box (BB) multi-horizon training loss with physical-consistency penalty:

$$\mathcal{L}_{\mathrm{BB}} = \sum_{k\in\{1,2,4\}} \frac{1}{N}\sum_i \big(\widehat{T}_i(t+k) - T_i(t+k)\big)^2 + \lambda_{\mathrm{phys}}\,\mathrm{ReLU}(|\Delta\widehat{T}| - \Delta T_{\max}). \tag{3}$$

Grey-box (GB) positive reparameterization of the zone capacitance:

$$C_{\mathrm{zon}} = c_s\big(\mathrm{softplus}(\rho) + c_{\min}/c_s\big), \qquad C_{\mathrm{zon}} \geq c_{\min} = 5\times 10^4\ \mathrm{J/K}. \tag{4}$$

Stage B maximum-a-posteriori identification of C_{zon} on the excitation subset $\mathcal{E}$ under a weak Gaussian prior:

$$\widehat{C}_{\mathrm{zon}} = \arg\min_C \sum_{i\in\mathcal{E}} \big(\widehat{T}_i(C) - T_i\big)^2 + \frac{1}{2\sigma_p^2}\big(C - C_{\mathrm{prior}}\big)^2. \tag{5}$$

Warm-started GB heat-term scaling: when a GB surrogate is warm-started from legacy BB weights, the inherited heat term is rescaled so that the per-step increment is dimensionally consistent at the runtime control step (the BB update lacks this scaling because its increment carries no explicit Δt):

$$s = s_{\text{legacy}} \frac{C_{\text{init}}}{\Delta t_{\text{legacy}}\, q_{\text{scale}}}. \tag{6}$$

Transfer-gap diagnostics: surrogate−live maintenance-score mismatch and mean action-gap norm:

$$\begin{aligned} \Delta m_s &= m_{s,\text{surrogate}} - m_{s,\text{BOPTEST}}, \\ g_a &= \frac{1}{N}\sum_{t=1}^{N} \left\| a_{t,\text{BOPTEST}} - a_{t,\text{surrogate}} \right\|_2. \end{aligned} \tag{7}$$

7 Additional tables cited from the main article

The following tables were moved here to reduce the float load of the main article; they are cited from it as Supplementary Tables.

Table S30: Maintenance-score decomposition across controller families (cited in the main article's controller-family synthesis). Thermostatic/HDRL are 14-day windows; MORL/PI are yearly means.

Controller (window/eval)	m_s	r_{time}	r_{sev}
pure BB (peak)	0.073	0.015	0.058
direct GB (peak)	1.046	0.771	0.276
hybrid λ_T=0.10 (peak)	0.087	0.047	0.040
HDRL λ_T=0 (peak)	0.180	0.061	0.119
pure BB (typical)	0.095	0.044	0.051
direct GB (typical)	1.102	0.824	0.278
hybrid λ_T=0.10 (typical)	0.041	0.024	0.017
HDRL λ_T=0 (typical)	0.234	0.031	0.202
MORL 50/50 (yearly, N=5)	0.187	0.130	0.057
MORL 75/25 (yearly, N=5)	0.139	0.092	0.047
PI (yearly)	0.910	0.636	0.274

Table S31: Controller-learning falsifiable-hypothesis ledger (CL1–CL5), cited in the main article's central fidelity–utility test. These are distinct from the paper-level hypotheses H1–H5: CL1–CL3 are the controller-experiment counterparts of H1–H3, and CL4–CL5 add MORL-specific checks. Each hypothesis is tested on the live BOPTEST RTE.

	Hypothesis	Evidence	Verdict
CL1	The higher-fidelity twin (GB) used directly as the rollout environment yields the best controller.	central test	FALSIFIED (direct GB $m_s > 1.0$)
CL2	Calibrated GB is useful when its role changes from dynamics provider to a frozen reward-shaping censor.	hybrid remedy	SUPPORTED (hybrid gives best cross-window robustness)
CL3	The thermostatic-optimal censor weight $\lambda_T = 0.10$ transfers to the HDRL family.	HDRL sweep	FALSIFIED (HDRL best at $\lambda_T = 0$)
CL4	MORL viability is determined by the reward scalarization alone.	MORL 5D/17D	FALSIFIED (needs the 17D observation)
CL5	The single-seed MORL canonical ($m_s \approx 0.10$) is representative.	MORL N=5	FALSIFIED at N=5 (best of five; CV 0.42–0.61)

Table S32: Modeling assumptions of the direct supply-temperature (T_{sup}) control interface (cited in the main article's predictive-validation section).

Assumption	Statement
Actuator abstraction	Supply-air temperature is commanded directly; valve/coil dynamics are not modelled beyond the Stage A latency term.
Command bounds	$T_{\text{sup}} \in [18, 35]\,°\text{C}$; fan command $u_{\text{fan}} \in [0, 1]$.
Power non-negativity	$\widehat{P} = \text{Softplus}(\cdot) \geq 0$, capped at $P_{\max} = 5500$ W.
State support	Zone and ambient temperatures normalized to $[-1, 1]$ over $[15, 35]$ and $[-10, 40]\,°\text{C}$.
Interface invariance	All four backends share the same transition signature; only the internal dynamics differ.

Table S33: Transition-dataset schema for the surrogate models. Every field of a training tuple (s_t, a_t, d_t, s_{t+1}) is tagged by role: **M** BOPTEST measurement, **X** exogenous disturbance, **D** derived feature, **A** control action, **T** regression target. In a recursive rollout the predicted targets update the physical-state block of the next observation ($\widehat{T}_{\text{zone},t+1} = T_{\text{zone},t} + \Delta\widehat{T}_{\text{zone},t+1}$, power, previous action), while exogenous, forecast, and time fields advance from the disturbance stream; the policy then emits a_{t+1} and the map f_θ is re-applied.

Field	Role	Description
$T_{\text{zone},t}$	M	zone air temperature (BOPTEST sensor); in the observation and, via ΔT, in the target
P_t (HVAC power)	M	measured heating/cooling power; observation and power-head target
CO_2	M	zone CO_2 concentration; observation only
$T_{\text{amb},t}$	X	ambient temperature (weather); observation
solar gain, occupancy, comfort set-points	X	exogenous drivers in d_t; affect the dynamics, not all directly observed
$\widehat{T}_{\text{amb},\,t+1:t+24\text{h}}$	D	5 ambient forecasts (+1, +3, +6, +12, +24 h); observation
cyclic time features	D	hour/day sine–cosine encodings; observation
prev. supply temp. $T_{\text{sup},t-1}$, prev. action a_{t-1}	D	control history; observation
$\Delta T_{\text{zone},t}$	D	causal-smoothed temperature delta; observation
$a_t = a_0$ (supply-T)	A	commanded supply-temperature set-point (main-text action map); the only deployed control input
a_1 (fan)	A	fan command; excited only during offline corpus generation, held at the BOPTEST baseline at deployment
$\Delta\widehat{T}_{\text{zone},t+1}$	T	predicted next-step zone-temperature increment
$\widehat{P}_{t+1}$	T	predicted next-step HVAC power (power head)
internal surrogate state (GB: RC state / C_{zon})	—	latent; propagated by the RC balance (GB) or carried within s_t (BB)

Table S34: Staged inverse calibration of the GB surrogate under the `block1_15min_episodeaware` preset; a second `power_head_only` preset freezes C_{zon} and refines only the power residual head (80 epochs).

Stage	Calibrates	Key result
A	telemetry alignment: one-step latency & bias	bias $-0.105\,°\text{C}$; post-fit temperature RMSE $0.238\,°\text{C}$
B	zone capacitance C_{zon} on the top-excitation subset (403/8058 rows, quantile 0.95; 120 epochs)	$C_{\text{zon}} = 4.413 \times 10^5$ J/K (+5.1% over prior); last-20-epoch stability $\pm 1{,}005$ J/K (0.23%); Laplace/Fisher $1\sigma = 6.6\%$, 95% CI $[3.845, 4.980] \times 10^5$ J/K ($n = 1838$)
C	neural residual heads, C_{zon} frozen (60 epochs)	refines residual fit while preserving the physical parameter

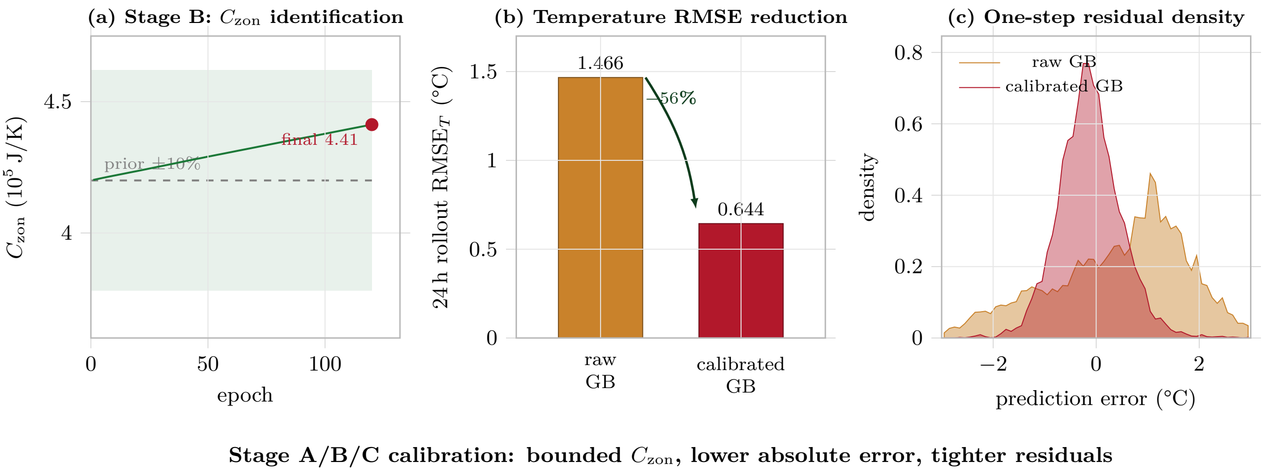


Figure S26: Stage A/B/C diagnostics: C_{zon} convergence, rollout temperature-error reduction, and residual tightening.

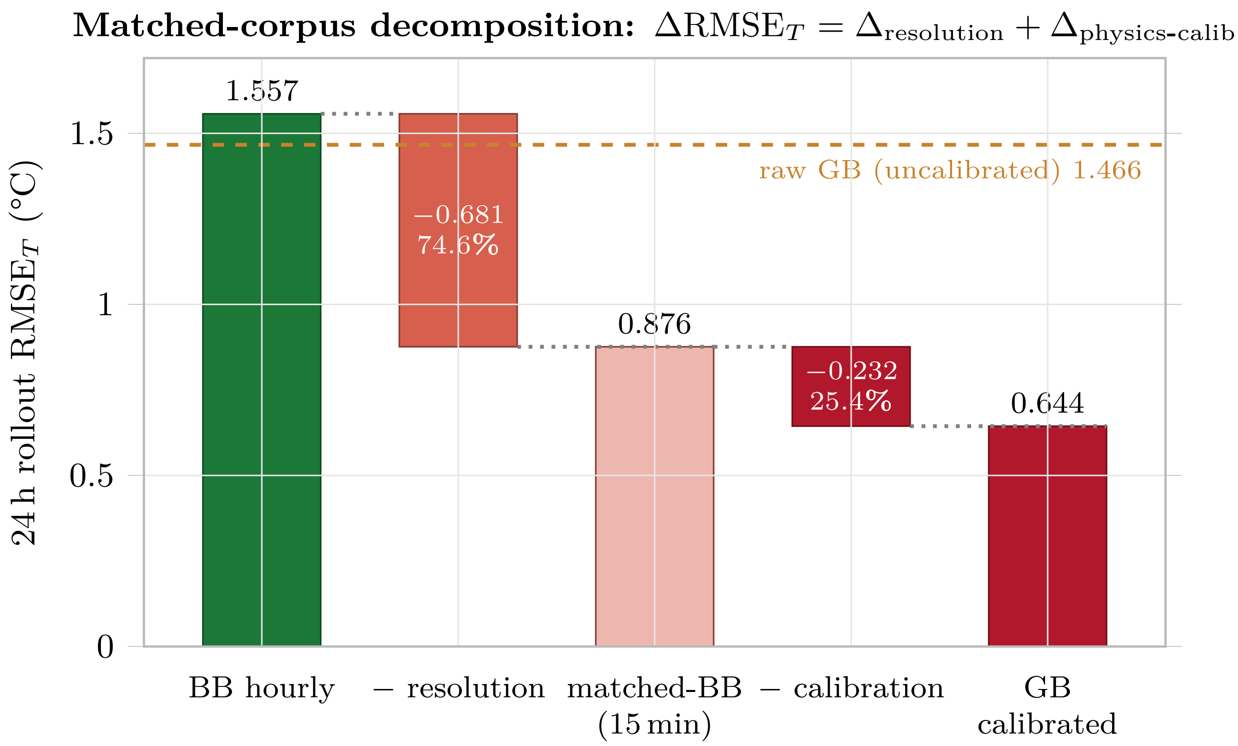
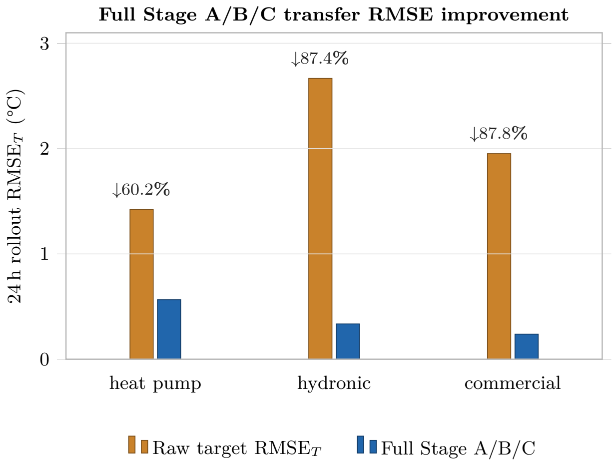
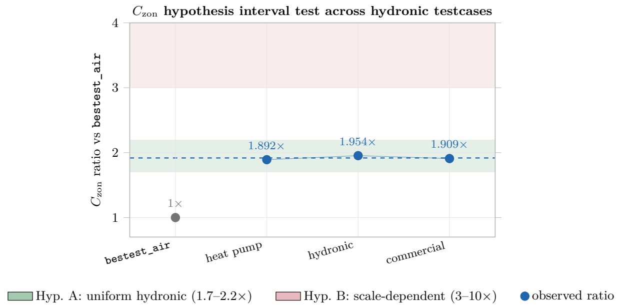


Figure S27: Matched-corpus attribution. The BB-to-GB gain is split into the 15-minute corpus contribution and the Stage A/B/C calibration contribution.



(a) Full Stage A/B/C RMSE improvement.



(b) C_{zon} hypothesis interval.

Figure S28: Surrogate-side transfer and physical consistency. Full Stage A/B/C reduces target rollout error on all hydronic-family testcases, while the re-identified C_{zon} ratios fall inside the uniform hydronic-family interval and outside the scale-dependent interval.

Table S35: Pre-specified hypothesis closure.

Hypothesis	Claim	Verdict	Evidence
T1 (strong)	Frozen mode=none recipe gives deployment-ready transfer.	FALSIFIED	Residential cases fail the 1.25× PI threshold; commercial is threshold-pass only with +35.3% energy.
T2 (medium)	Partial Stage C recovers controller transfer.	FALSIFIED (structural)	Partial recalibrates only the surrogate; the controller is frozen by scope, so live KPI cannot change.
T3 (surrogate)	Full Stage A/B/C gives a usable target surrogate.	SUPPORTED (N=3)	RMSE$_T$ gains 60.2%, 87.4%, 87.8%; C_{zon} re-identified consistently.
T3 (controller)	Frozen controller transfers given a valid adapter.	FALSIFIED	Residential cases fail comfort; commercial passes safety but overuses energy, regime-dependent failure.
hierarchy	Verdict is monotone in recalibration depth.	SUPPORTED	Controller KPI invariant under surrogate-only recalibration; surrogate fidelity improves with full recalibration.

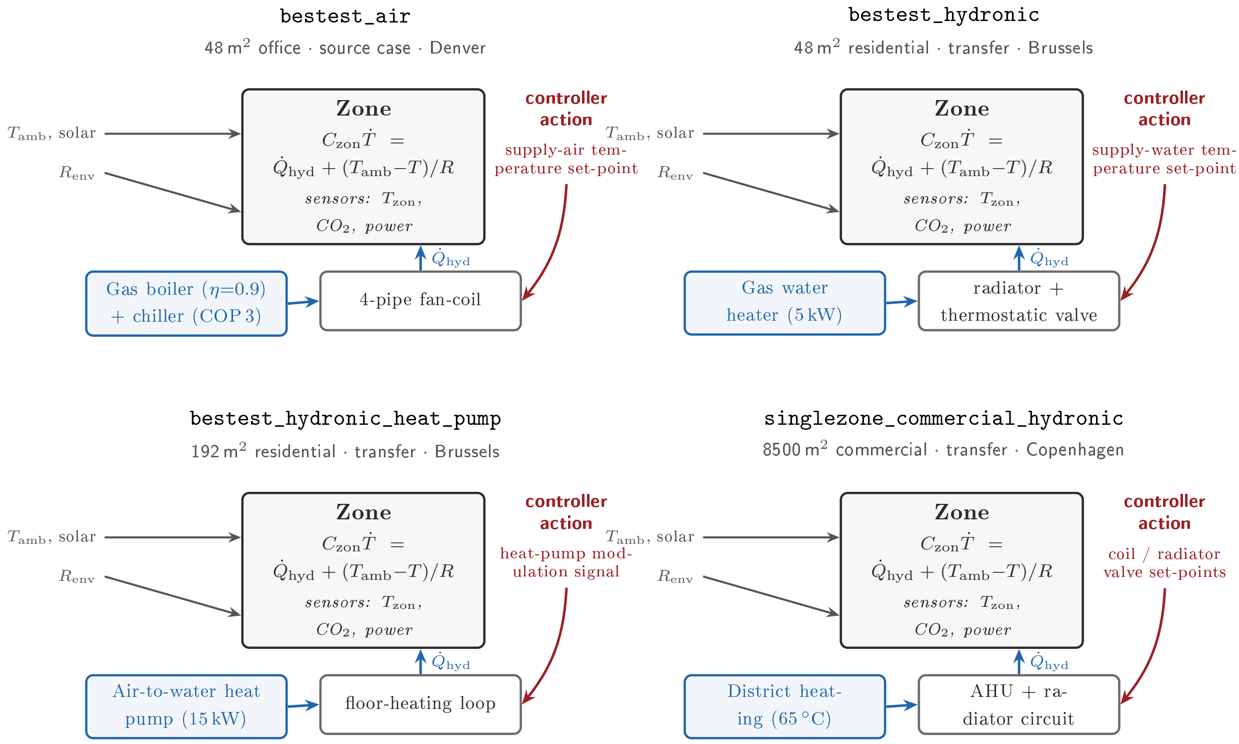


Figure S29: Schematics of the four BOPTEST test cases (source `bestest_air` plus the three hydronic transfer targets), from the official BOPTEST documentation. All share the single-zone energy balance but differ in heat source, distribution path, and actuator interface; the highlighted arrow marks the scalar command channel or adapter-mediated actuator interface used by the controller in each case.